

PC Build Store

OOP Project

Final Submission

Group Members

- Abdulhanan - Dashboard, Reports, AI Chat
- Ahmad Mushtaq (2501803) - Billing Module
- Aitzaz (2501805) - Build Upgrades
- Haroon - Build Configurator

Object Oriented Programming - Final Submission

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1. Project Overview

A PC part picker and configurator built with Java 17, Swing GUI, and MariaDB. Users can browse 78+ components across 6 categories, build custom PCs with compatibility filtering, upgrade existing builds, purchase builds with receipts, and get AI-powered build suggestions via an integrated chat bot.

2. Tech Stack

Component	Technology
Language	Java 17
GUI	Swing
Database	MariaDB 10.4 (XAMPP)
Connectivity	JDBC (mysql-connector-j-9.7.0)
Build System	NetBeans Ant
AI Backend	KoboldCPP (OpenAI-compatible API)

3. Setup Instructions

Step 1
Install XAMPP and start MySQL service
Step 2
Import database: <code>mysql -u root < Database/pc_build_store.sql</code>
Step 3
Open project in NetBeans or build with: <code>ant compile</code>
Step 4
Run application: <code>ant run</code>

4. Submission Structure

Folder	Contents
Source_Code/	28 Java source files (com.pcbuildstore package)
Database/	SQL schema and seed data
Report/	Project report (PDF)
Analysis/	Architecture and ER diagram
Contribution_Details/	Individual contribution records

Output_Screenshots/	Application screenshots
Presentation/	Presentation materials

5. GitHub Repository

<https://github.com/Abdulhanan535/PCBuildStore>